ANSWER KEY.

Q1. $\int \frac{3x+1}{x^2+2x+5}\,dx = \int \frac{3x+3-2}{(x+1)^2+4}\,dx = \int \frac{3(x+1)}{(x+1)^2+4}\,dx - \int \frac{2}{(x+1)^2+4}\,dx$

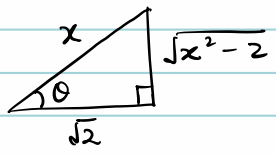
$\cdots \equiv \frac{3}{2}\ln\left|(x+1)^2+4\right| - \frac{2}{2}\arctan\left(\frac{x+1}{2}\right) + C$

$= \frac{3}{2}\ln\left(x^2+2x+5\right) - \arctan\left(\frac{x+1}{2}\right) + C$

Q2. $\int \frac{1}{x^2\sqrt{x^2-2}}\,dx \overset{x=\sqrt{2}\sec\theta}{=\!=} \int \frac{1}{2\sec^2\theta\sqrt{2\sec^2\theta-2}}\sqrt{2}\sec\theta\tan\theta\,d\theta$

$= \frac{\sqrt{2}}{2\sqrt{2}}\int \frac{\cos^2\theta}{\tan\theta}\sec\theta\tan\theta\,d\theta = \frac{1}{2}\int \cos\theta\,d\theta = \frac{1}{2}\sin\theta + C$

$= \frac{1}{2}\sin\left(\sec^{-1}\left(\frac{x}{\sqrt{2}}\right)\right) + C = \frac{1}{2}\frac{\sqrt{x^2-2}}{x} + C.$



Q3. $\int \frac{x^2}{(x-1)(x^2+1)}\,dx =: I$ (:= means "define")

First notice $2I$

$= \int \frac{2x^2}{(x-1)(x^2+1)}\,dx = \int \frac{x^2+1+x^2-x+x-1}{(x-1)(x^2+1)}\,dx = \int \frac{1}{x-1}\,dx + \int \frac{x}{x^2+1}\,dx + \int \frac{1}{x^2+1}\,dx$

$= \ln|x-1| + \frac{1}{2}\ln(x^2+1) + \arctan x + C.$

Thus

$I = \frac{2I}{2} = \frac{1}{2}\ln|x-1| + \frac{1}{4}\ln(x^2+1) + \frac{1}{2}\arctan x + C.$

Q4. $\int \frac{1}{1+\sin x}\,dx = \int \frac{1}{1+\sin x}\frac{1-\sin x}{1-\sin x}\,dx = \int \frac{1-\sin x}{1-\sin^2 x}\,dx$

$= \int \frac{1}{\cos^2 x}\,dx + \int \frac{-\sin x}{\cos^2 x}\,dx = \int \sec^2 x\,dx + \int \frac{1}{u^2}\,du \quad (u=\cos x)$

$= \tan x - \frac{1}{\cos x} + C = \tan x - \sec x + C.$

Q5. $\int \ln(1+x^2)\,dx$

IBP (IPP): $u = \ln(1+x^2)$, $dv = dx$

$du = \frac{1}{1+x^2}\cdot 2x\,dx$, $v = x$

$= \ln(1+x^2)\cdot x - \int \frac{2x^2}{1+x^2}\,dx$

$= x\ln(1+x^2) - \int \frac{2x^2+2}{1+x^2}\,dx + \int \frac{2}{1+x^2}\,dx$

$= x\ln(1+x^2) - 2x + 2\arctan x + C$.

Q6. $\int \frac{\sqrt{x}}{1+x}\,dx \overset{u=\sqrt{x}}{=\!=\!=} \int \frac{u}{1+u^2}\cdot 2u\,du = \int \frac{2u^2+2}{u^2+1}\,du - \int \frac{2}{u^2+1}\,du$

$= 2u - 2\arctan(u) + C = 2\sqrt{x} - 2\arctan(\sqrt{x}) + C$.

Q7. $\int \sec^3 x \tan^4 x\,dx =: I$.

$I = \int \sec^3 x\,(\sec^2 x - 1)^2\,dx = \int \sec^7 x\,dx - 2\int \sec^5 x\,dx + \int \sec^3 x\,dx$.

$(dv = \sec^2 x\,dx)$ $\qquad$ I_7 $\qquad$ I_5 $\qquad$ I_3

$I_7 = \int \sec^7 x\,dx = \sec^5 x\tan x - \int 5\sec^4 x\cdot \sec x\tan x\cdot \tan x\,dx$

$= \sec^5 x\tan x - 5\int \underset{I_7}{\sec^7 x}\,dx + 5\int \sec^5 x\,dx$

$\Rightarrow I_7 = \frac{\sec^5 x\tan x}{6} + \frac{5}{6} I_5$.

$I_5 = \int \sec^5 x\,dx = \sec^3 x\tan x - \int 3\sec^2 x\cdot \sec x\tan x\cdot \tan x\,dx$

$= \sec^3 x\tan x - 3\int \sec^5 x\,dx + 3\int \sec^3 x\,dx$

$\Rightarrow I_5 = \frac{\sec^3 x\tan x}{4} + \frac{3}{4} I_3$

$$I_3 = \int \sec^3 x\,dx = \sec x \tan x - \int \sec x \tan x \tan x\,dx$$

$$= \sec x \tan x - \int \sec^3 x\,dx + \int \sec x\,dx$$

$$\Rightarrow I_3 = \frac{\sec x \tan x}{2} + \frac{1}{2}\int \sec x \frac{\tan x + \sec x}{\tan x + \sec x}\,dx$$

$$= \frac{\sec x \tan x}{2} + \frac{\ln|\tan x + \sec x|}{2} \quad (+C).$$

$$I = I_7 - 2I_5 + I_3 = \cdots$$

$$= \frac{\sec^5 x \tan x}{6} - \frac{7}{24}\sec^3 x \tan x + \frac{\sec x \tan x}{16} + \frac{\ln|\tan x + \sec x|}{16} + C.$$

Q8. $$\int \frac{x+1}{x^2+x+1}\,dx = \int \frac{x+\frac{1}{2}}{(x+\frac{1}{2})^2+\frac{3}{4}}\,dx + \int \frac{1/2}{(x+\frac{1}{2})^2+\frac{3}{4}}\,dx$$

$$= \frac{1}{2}\ln\left|(x+\tfrac{1}{2})^2+\tfrac{3}{4}\right| + \frac{1}{2}\frac{2}{\sqrt{3}}\arctan\left(\frac{x+\frac{1}{2}}{\frac{\sqrt{3}}{2}}\right) + C$$

$$= \frac{1}{2}\ln(x^2+x+1) + \frac{1}{\sqrt{3}}\arctan\left(\frac{2x+1}{\sqrt{3}}\right) + C.$$

Q9. $$\int \frac{e^{2x}}{e^x+1}\,dx \overset{u=e^x+1}{=\!=} \int \frac{(u-1)^2}{u}\cdot\frac{1}{u-1}\,du = \int \frac{u-1}{u}\,du = u - \ln|u| + C$$

$$= e^x + 1 - \ln(e^x+1) + C$$

$$(= e^x - \ln(e^x+1) + C.)$$

Q10. $$\int x \arctan x\,dx = \frac{x^2}{2}\arctan x - \int \frac{x^2}{2}\frac{1}{1+x^2}\,dx \qquad (dv = x\,dx)$$

$$= \frac{x^2\arctan x}{2} - \frac{1}{2}\int \frac{1+x^2}{1+x^2}\,dx + \frac{1}{2}\int \frac{1}{1+x^2}\,dx$$

$$= \frac{x^2\arctan x}{2} - \frac{1}{2}x + \frac{1}{2}\arctan(x) + C$$

— LEVEL UP —

Q11. $\int \frac{1}{x^4+1}\,dx$.

[#1 - Somewhat Brute force]

$$\int \frac{1}{x^4+1}\,dx = \int \frac{1}{x^4+2x^2+1-2x^2}\,dx = \int \frac{1}{(x^2+1-\sqrt{2}x)(x^2+1+\sqrt{2}x)}\,dx$$

$$= \int \frac{(x^2+\sqrt{2}x+1)\left(-\frac{1}{2\sqrt{2}}x+\frac{1}{2}\right) + (x^2-\sqrt{2}x+1)\left(\frac{1}{2\sqrt{2}}x+\frac{1}{2}\right)}{(x^2-\sqrt{2}x+1)(x^2+\sqrt{2}x+1)}\,dx$$

/* Google "partial fraction decomposition" */

$$= \frac{1}{4\sqrt{2}}\left(\int \frac{-2x+2\sqrt{2}}{x^2-\sqrt{2}x+1}\,dx + \int \frac{2x+2\sqrt{2}}{x^2+\sqrt{2}x+1}\,dx\right)$$

$$= \frac{1}{4\sqrt{2}}\left(\int \frac{-2x+\sqrt{2}}{x^2-\sqrt{2}x+1}\,dx + \int \frac{\sqrt{2}}{x^2-\sqrt{2}x+1}\,dx + \int \frac{2x+\sqrt{2}}{x^2+\sqrt{2}x+1}\,dx + \int \frac{\sqrt{2}}{x^2+\sqrt{2}x+1}\,dx\right)$$

$$= \frac{1}{4\sqrt{2}}\left[-\ln|x^2-\sqrt{2}x+1| + \ln|x^2+\sqrt{2}x+1| + \sqrt{2}\left(\int \frac{1}{(x-\frac{1}{\sqrt{2}})^2+\frac{1}{2}}\,dx + \int \frac{1}{(x+\frac{1}{\sqrt{2}})^2+\frac{1}{2}}\,dx\right)\right]$$

$x^2-\sqrt{2}x+1 \geq 0$, $x^2+\sqrt{2}x+1 \geq 0$ — can you see why?

$$= \frac{1}{4\sqrt{2}}\left[\ln\left(\frac{x^2+\sqrt{2}x+1}{x^2-\sqrt{2}x+1}\right) + \sqrt{2}\cdot\left(\frac{1}{\frac{1}{\sqrt{2}}}\arctan\left(\frac{x-\frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{2}}}\right) + \frac{1}{\frac{1}{\sqrt{2}}}\arctan\left(\frac{x+\frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{2}}}\right)\right)\right] + C$$

$$= \frac{1}{4\sqrt{2}}\ln\left(\frac{x^2+\sqrt{2}x+1}{x^2-\sqrt{2}x+1}\right) + \frac{1}{2\sqrt{2}}\arctan(\sqrt{2}x-1) + \frac{1}{2\sqrt{2}}\arctan(\sqrt{2}x+1) + C.$$

[#2 - Extreme Cleverness - Symmetry + completing the square]

$$\int \frac{1}{x^4+1}\,dx = \int \frac{\frac{1}{2}(1+x^2+1-x^2)}{1+x^4}\,dx = \frac{1}{2}\int \frac{1+x^2}{1+x^4}\,dx + \frac{1}{2}\int \frac{1-x^2}{1+x^4}\,dx$$

$$= \frac{1}{2}\int \frac{\frac{1}{x^2}+1}{x^2+\frac{1}{x^2}}\,dx + \frac{1}{2}\int \frac{\frac{1}{x^2}-1}{x^2+\frac{1}{x^2}}\,dx$$

$$= \frac{1}{2}\int \frac{\frac{1}{x^2}+1}{(x-\frac{1}{x})^2+2}\,dx + \frac{1}{2}\int \frac{\frac{1}{x^2}-1}{(x+\frac{1}{x})^2-(\sqrt{2})^2}\,dx$$

$u = x - \frac{1}{x}$ $\qquad$ $v = x + \frac{1}{x}$

$$= \frac{1}{2}\int \frac{1}{u^2+2}\,du + \frac{1}{2}\int \frac{-1}{v^2-(\sqrt{2})^2}\,dv$$

$$= \frac{1}{2\sqrt{2}} \arctan\left(\frac{u}{\sqrt{2}}\right) - \frac{1}{2}\int \frac{v+\sqrt{2}-(v-\sqrt{2})}{(v-\sqrt{2})(v+\sqrt{2})}\cdot\frac{1}{2\sqrt{2}}\,dv$$

$$= \frac{1}{2\sqrt{2}} \arctan\left(\frac{x-\frac{1}{x}}{\sqrt{2}}\right) - \frac{1}{4\sqrt{2}}\int \frac{1}{v-\sqrt{2}}\,dv + \frac{1}{4\sqrt{2}}\int \frac{1}{v+\sqrt{2}}\,dv$$

$$= \frac{1}{2\sqrt{2}} \arctan\left(\frac{x-\frac{1}{x}}{\sqrt{2}}\right) - \frac{1}{4\sqrt{2}}\ln|v-\sqrt{2}| + \frac{1}{4\sqrt{2}}\ln|v+\sqrt{2}| + C$$

$$= \frac{1}{2\sqrt{2}} \arctan\left(\frac{x-\frac{1}{x}}{\sqrt{2}}\right) - \frac{1}{4\sqrt{2}}\ln\left|x+\frac{1}{x}-\sqrt{2}\right| + \frac{1}{4\sqrt{2}}\ln\left|x+\frac{1}{x}+\sqrt{2}\right| + C$$

$$= \frac{1}{2\sqrt{2}} \arctan\left(\frac{x-\frac{1}{x}}{\sqrt{2}}\right) + \frac{1}{4\sqrt{2}}\ln\left|\frac{x+\frac{1}{x}+\sqrt{2}}{x+\frac{1}{x}-\sqrt{2}}\right| + C$$

$$= \frac{1}{2\sqrt{2}} \arctan\left(\frac{x-\frac{1}{x}}{\sqrt{2}}\right) + \frac{1}{4\sqrt{2}}\ln\left(\frac{x^2+\sqrt{2}x+1}{x^2-\sqrt{2}x+1}\right) + C$$

Q 12. $\int \frac{1}{\sqrt{e^x+1}}\,dx \overset{u=\sqrt{e^x+1}}{=\!=\!=} \int \frac{2}{u^2-1}\,du = \int \frac{u+1-(u-1)}{(u-1)(u+1)}\,du = \int \frac{1}{u-1}\,du - \int \frac{1}{u+1}\,du$

$$= \ln\left|\frac{u-1}{u+1}\right| + C = \ln\frac{\sqrt{e^x+1}-1}{\sqrt{e^x+1}+1} + C.$$

Q 13. $\int_0^{\pi/2} \frac{\sin x}{\cos x + \sin x}\,dx$ (this technique is known as "King's rule")

$$\overset{u=\pi/2-x}{=\!=\!=} \int_{\pi/2}^{0} \frac{\sin(\pi/2-u)}{\cos(\pi/2-u)+\sin(\pi/2-u)}(-du) = \int_0^{\pi/2} \frac{\cos u}{\sin u+\cos u}\,du = \int_0^{\pi/2} \frac{\cos x}{\sin x+\cos x}\,dx$$

$$\Rightarrow 2\int_0^{\pi/2} \frac{\sin x}{\cos x+\sin x}\,dx = \int_0^{\pi/2} \frac{\sin x}{\cos x+\sin x}\,dx + \int_0^{\pi/2} \frac{\cos x}{\cos x+\sin x}\,dx = \int_0^{\pi/2} \frac{\cos x+\sin x}{\cos x+\sin x}\,dx$$

$$= \int_0^{\pi/2} 1\,dx = \pi/2$$

$$\Rightarrow \int_0^{\pi/2} \frac{\sin x}{\cos x+\sin x}\,dx = \frac{\pi/2}{2} = \pi/4.$$

Q 14. $\int \frac{x \arctan x}{(1+x^2)^2}\,dx \overset{t=\arctan x}{=\!=\!=} \int \frac{t\cdot\tan t}{1+\tan^2 t}\,dt = \int t\,\frac{\sin t}{\cos t}\cos^2 t\,dt$

$$= \int t \sin t \cos t \, dt = \int \frac{1}{2} t \sin(2t) \, dt$$

$u = t$, $dv = \sin(2t)\,dt$

$du = dt$, $v = -\frac{1}{2}\cos(2t)$

$$= \frac{1}{2}\left(-\frac{1}{2} t\cos(2t) + \frac{1}{2}\int \cos(2t)\,dt\right)$$

$$= -\frac{1}{4} t\cos(2t) + \frac{1}{4}\cdot\frac{1}{2}\sin(2t) + C$$

$$= -\frac{1}{4} t(2\cos^2 t - 1) + \frac{1}{4}\sin t\cos t + C$$

$$= -\frac{1}{2}\frac{t}{\sec^2 t} + \frac{1}{4}t + \frac{1}{4}\frac{\sin t}{\cos t}\cdot\cos^2 t + C$$

$$= -\frac{1}{2}\frac{t}{1+\tan^2 t} + \frac{1}{4}t + \frac{1}{4}\frac{\tan t}{1+\tan^2 t} + C$$

$$= -\frac{1}{2}\frac{\arctan x}{1+x^2} + \frac{1}{4}\arctan x + \frac{1}{4}\frac{x}{1+x^2} + C.$$

Q15. $\int_0^1 \frac{\ln(1+x)}{1+x^2}\,dx \overset{x=\tan\theta}{=\!=} \int_0^{\pi/4} \frac{\ln(1+\tan\theta)}{1+\tan^2\theta}\sec^2\theta\,d\theta = \int_0^{\pi/4} \ln(1+\tan\theta)\,d\theta$

$= I$

$$= \int_{\pi/4}^{0} \ln\left(1+\tan\left(\pi/4-\theta\right)\right)(-d\theta) \quad \text{(King's rule again)}$$

$$= \int_0^{\pi/4} \ln\left(1+\frac{1-\tan\theta}{1+\tan\theta}\right)d\theta = \int_0^{\pi/4}\left[\ln 2 - \ln(1+\tan\theta)\right]d\theta$$

$$= \frac{\ln 2}{4}\pi - \int_0^{\pi/4} \ln(1+\tan\theta)\,d\theta$$

$$\Rightarrow \quad I = \frac{\ln 2}{4}\pi - I \quad \Rightarrow \quad I = \frac{\ln 2}{8}\pi.$$

Q16. $\int \frac{1}{\sqrt{x}+\sqrt[3]{x}}\,dx \overset{x=t^6}{=\!=} \int \frac{1}{t^3+t^2} 6t^5\,dt = 6\int \frac{t^3}{t+1}\,dt$

$$= 6\left(\int \frac{t^3+1}{t+1}\,dt - \int \frac{1}{t+1}\,dt\right) = 6\int (t^2-t+1)\,dt - 6\ln|t+1|$$

$$= 2t^3 - 3t^2 + 6t - 6\ln|t+1| + C$$

$$= 2x^{\frac{1}{2}} - 3x^{\frac{1}{3}} + 6x^{\frac{1}{6}} - 6\ln\left(x^{\frac{1}{6}}+1\right) + C$$

Q17. $\int \frac{\sin x}{\cos x + \sin x + 1}\, dx$

This is an advanced trick called the Weierstrass substitution, usually (a last resort) for rational trig integrals.

$$= \int \frac{\sin\left(2\cdot\frac{x}{2}\right)}{\cos\left(2\cdot\frac{x}{2}\right) + \sin\left(2\cdot\frac{x}{2}\right) + 1}\, dx = \int \frac{2\sin\left(\frac{x}{2}\right)\cos\left(\frac{x}{2}\right)}{2\cos^2\left(\frac{x}{2}\right) - 1 + 2\sin\left(\frac{x}{2}\right)\cos\left(\frac{x}{2}\right) + 1}\, dx$$

$$= \int \frac{2\tan\left(\frac{x}{2}\right)}{2 + 2\tan\left(\frac{x}{2}\right)}\, dx \quad \boxed{t = \tan\left(\frac{x}{2}\right)} \quad \int \frac{t}{1+t}\cdot\frac{2}{1+t^2}\, dt$$

$$= \int \frac{2}{1+t^2}\, dt - \int \frac{2}{(1+t)(1+t^2)}\, dt = 2\arctan(t) - \int \frac{(1+t^2) - (t-1)(1+t)}{(1+t)(1+t^2)}\, dt$$

$$= 2\arctan\left(\tan\left(\tfrac{x}{2}\right)\right) - \int \frac{1}{1+t}\, dt + \int \frac{t-1}{t^2+1}\, dt$$

$$= x - \ln|1+t| + \tfrac{1}{2}\ln|t^2+1| - \arctan(t) + C$$

$$= x - \ln\left|1+\tan\tfrac{x}{2}\right| + \tfrac{1}{2}\ln\left(\sec^2\tfrac{x}{2}\right) - \tfrac{x}{2} + C$$

$$= \tfrac{x}{2} + \ln\left|\frac{\sec\left(\frac{x}{2}\right)}{1+\tan\frac{x}{2}}\right| + C$$

[Optional: $= \frac{x}{2} + \ln\frac{1}{\left|\cos\frac{x}{2} + \sin\frac{x}{2}\right|} + C = \frac{x}{2} - \ln\left|\cos\tfrac{x}{2} + \sin\tfrac{x}{2}\right| + C.$]

Q18. Note that you may also prove this formally using mathematical induction.

$$\int_0^1 x^m (1-x)^n\, dx \qquad \begin{vmatrix} u = (1-x)^n \\ dv = x^m\, dx \end{vmatrix}$$

$$= \frac{1}{m+1} x^{m+1}(1-x)^n \Big|_0^1 - \int_0^1 (-n)(1-x)^{n-1}\frac{1}{m+1}x^{m+1}\, dx$$

$$= 0 + \frac{n}{m+1}\int_0^1 (1-x)^{n-1} x^{m+1}\, dx$$

$$= \frac{n}{m+1}\left(\frac{n-1}{m+2}\int_0^1 (1-x)^{n-2}x^{m+2}dx\right)$$

$$= \cdots = \frac{n}{m+1}\frac{n-1}{m+2}\cdots\frac{n-j}{m+j+1}\int_0^1 (1-x)^{n-j-1}x^{m+j+1}dx$$

$$= \frac{n}{m+1}\frac{n-1}{m+2}\cdots\frac{n-(n-1)}{m+(n-1)+1}\int_0^1 (1-x)^{0}x^{m+(n-1)+1}dx$$

$$= \frac{n!}{\frac{(m+n)!}{m!}}\int_0^1 x^{m+n}dx = \frac{n!\,m!}{(m+n+1)!}x^{m+n+1}\Big|_0^1 = \frac{n!\,m!}{(m+n+1)!}.$$

Q19. $$\int \frac{1}{1+x^3}dx = \int \frac{1}{(1+x)(1-x+x^2)}dx = \int \frac{(1-x+x^2)-(x-2)(x+1)}{(1+x)(1-x+x^2)}\cdot\frac{1}{3}dx$$

$$= \frac{1}{3}\left[\int \frac{1}{1+x}dx - \int \frac{x-2}{1-x+x^2}dx\right]$$

$$= \frac{1}{3}\ln|1+x| - \frac{1}{3}\left(\int \frac{2x-1}{1-x+x^2}\cdot\frac{1}{2}dx - \int \frac{\frac{3}{2}}{(x-\frac{1}{2})^2+\frac{3}{4}}dx\right)$$

$$= \frac{1}{3}\ln|1+x| - \frac{1}{6}\ln|1-x+x^2| + \frac{1}{2}\cdot\frac{1}{\frac{\sqrt{3}}{2}}\arctan\left(\frac{x-\frac{1}{2}}{\frac{\sqrt{3}}{2}}\right)+C$$

$$= \frac{1}{3}\ln|1+x| - \frac{1}{6}\ln(1-x+x^2) + \frac{1}{\sqrt{3}}\arctan\left(\frac{2x-1}{\sqrt{3}}\right)+C.$$

Q20. $$\int \frac{x^4+1}{x^6+1}dx = \int \frac{\frac{2}{3}(x^4-x^2+1)+(\frac{1}{3}x^2+\frac{1}{3})(x^2+1)}{(x^2+1)(x^4-x^2+1)}dx$$

$$= \frac{2}{3}\int\frac{1}{x^2+1}dx + \frac{1}{3}\int\frac{x^2+1}{x^4-x^2+1}dx = \frac{2}{3}\arctan(x) + \frac{1}{3}\int\frac{1+\frac{1}{x^2}}{x^2-1+\frac{1}{x^2}}dx$$

$$= \frac{2}{3}\arctan(x) + \frac{1}{3}\int\frac{(x-\frac{1}{x})'}{(x-\frac{1}{x})^2+1}dx = \frac{2}{3}\arctan x + \frac{1}{3}\arctan(x-\frac{1}{x})+C.$$

Q21. $$\int \frac{x^2+1}{x^4+x^2+1}dx$$

[#1] $$= \int \frac{1+\frac{1}{x^2}}{x^2+1+\frac{1}{x^2}}dx = \int \frac{1+\frac{1}{x^2}}{(x-\frac{1}{x})^2+3}dx = \frac{1}{\sqrt{3}}\arctan\left(\frac{x-\frac{1}{x}}{\sqrt{3}}\right)+C$$

[#2]

$$\int \frac{x^2+1}{x^4+x^2+1}dx = \int \frac{x^2+1}{(x^2+1)^2-x^2}dx = \int \frac{x^2+1}{(x^2-x+1)(x^2+x+1)}dx$$

$$= \int \frac{\frac{1}{2}(x^2-x+1) + \frac{1}{2}(x^2+x+1)}{(x^2-x+1)(x^2+x+1)}\,dx = \frac{1}{2}\int \frac{1}{(x+\frac{1}{2})^2+\frac{3}{4}}\,dx + \frac{1}{2}\int \frac{1}{(x-\frac{1}{2})^2+\frac{3}{4}}\,dx$$

$$= \frac{1}{2}\cdot\frac{1}{\frac{\sqrt{3}}{2}}\arctan\left(\frac{x+\frac{1}{2}}{\frac{\sqrt{3}}{2}}\right) + \frac{1}{2}\cdot\frac{1}{\frac{\sqrt{3}}{2}}\arctan\left(\frac{x-\frac{1}{2}}{\frac{\sqrt{3}}{2}}\right) + C$$

$$= \frac{1}{\sqrt{3}}\arctan\left(\frac{2x+1}{\sqrt{3}}\right) + \frac{1}{\sqrt{3}}\arctan\left(\frac{2x-1}{\sqrt{3}}\right) + C.$$

$$\left[= \frac{1}{\sqrt{3}}\arctan\left(\frac{\frac{2x+1}{\sqrt{3}} + \frac{2x-1}{\sqrt{3}}}{1 - \frac{2x+1}{\sqrt{3}}\cdot\frac{2x-1}{\sqrt{3}}}\right) = \frac{1}{\sqrt{3}}\arctan\left(\frac{\sqrt{3}x}{1-x^2}\right)\right.$$

$$= \frac{1}{\sqrt{3}}\left(\frac{\pi}{2} - \arctan\left(\frac{1-x^2}{\sqrt{3}x}\right)\right) = \frac{\pi}{2\sqrt{3}} + \frac{1}{\sqrt{3}}\arctan\left(-\frac{1-x^2}{\sqrt{3}x}\right)$$

$$\left. = C' + \frac{1}{\sqrt{3}}\operatorname{arc}\left(\frac{x-\frac{1}{x}}{\sqrt{3}}\right). \right]$$

Q22. $\int \frac{1}{x(x^4+1)}\,dx$

$$= \int \frac{(x^4+1) + (-x^3)x}{x(x^4+1)}\,dx = \int \frac{1}{x}\,dx - \int \frac{x^3}{x^4+1}\,dx$$

$$= \int \frac{1}{x}\,dx - \int \frac{x^3}{(x^2+\sqrt{2}x+1)(x^2-\sqrt{2}x+1)}\,dx$$

$$= \ln|x| - \int \frac{(\frac{1}{2}x - \frac{1}{2\sqrt{2}})(x^2+\sqrt{2}x+1) + (\frac{1}{2}x + \frac{1}{2\sqrt{2}})(x^2-\sqrt{2}x+1)}{(x^2+\sqrt{2}x+1)(x^2-\sqrt{2}x+1)}\,dx$$

$$= \ln|x| - \left[\int \frac{\frac{1}{2}x - \frac{1}{2\sqrt{2}}}{x^2-\sqrt{2}x+1}\,dx + \int \frac{\frac{1}{2}x + \frac{1}{2\sqrt{2}}}{x^2+\sqrt{2}x+1}\,dx\right]$$

$$= \ln|x| - \frac{1}{4}\left[\int \frac{2x-\sqrt{2}}{x^2-\sqrt{2}x+1}\,dx + \int \frac{2x+\sqrt{2}}{x^2+\sqrt{2}x+1}\,dx\right]$$

$$= \ln|x| - \frac{1}{4}\left[\ln|x^2-\sqrt{2}x+1| + \ln|x^2+\sqrt{2}x+1|\right] + C$$

$$= \frac{1}{4}\ln\left(\frac{x^4}{x^4+1}\right) + C.$$

Q23. $\int e^{\sqrt{x}}\,dx \overset{u=\sqrt{x}}{=\!=} 2\int e^u u\,du \overset{\text{IBP}}{=\!=} 2(e^u u - u) + C = 2(e^{\sqrt{x}}\sqrt{x} - \sqrt{x}) + C.$

Q24. $\int \sqrt{\tan x}\,dx \overset{u=\sqrt{\tan x}}{=\!=} \int \frac{2u^2}{1+u^4}\,du = \int \frac{2}{\frac{1}{u^2}+u^2}\,du$

$$= \int \frac{1+\frac{1}{u^2}}{\frac{1}{u^2}+u^2}\,du + \int \frac{1-\frac{1}{u^2}}{\frac{1}{u^2}+u^2}\,du$$

$$= \int \frac{(u-\frac{1}{u})'}{(u-\frac{1}{u})^2+2}\,du + \int \frac{u^2}{1+u^4}\,du - \boxed{\int \frac{1}{1+u^4}\,du} \quad \text{Q11}$$

$$= \frac{1}{\sqrt{2}}\arctan\left(\frac{u-\frac{1}{u}}{\sqrt{2}}\right) - \left[\frac{1}{2\sqrt{2}}\arctan\left(\frac{u-\frac{1}{u}}{\sqrt{2}}\right) + \frac{1}{4\sqrt{2}}\ln\left(\frac{u^2+\sqrt{2}u+1}{u^2-\sqrt{2}u+1}\right)\right] + \int \frac{u^2}{1+u^4}\,du.$$

$$\Rightarrow \int \frac{u^2}{1+u^4}\,du = \frac{1}{2\sqrt{2}}\arctan\left(\frac{u-\frac{1}{u}}{\sqrt{2}}\right) - \frac{1}{4\sqrt{2}}\ln\left(\frac{u^2+\sqrt{2}u+1}{u^2-\sqrt{2}u+1}\right) + C$$

$$\Rightarrow \int \sqrt{\tan x}\,dx = \int \frac{2u^2}{1+u^4}\,du = \frac{1}{\sqrt{2}}\arctan\left(\frac{u-\frac{1}{u}}{\sqrt{2}}\right) - \frac{1}{2\sqrt{2}}\ln\left(\frac{u^2+\sqrt{2}u+1}{u^2-\sqrt{2}u+1}\right) + C$$

$$= \frac{1}{\sqrt{2}}\arctan\left(\frac{\sqrt{\tan x}-\frac{1}{\sqrt{\tan x}}}{\sqrt{2}}\right) - \frac{1}{2\sqrt{2}}\ln\left(\frac{\tan x+\sqrt{2}\sqrt{\tan x}+1}{\tan x-\sqrt{2}\sqrt{\tan x}+1}\right) + C$$

Q25. $$\int \frac{x^2-1}{x^4+3x^2+1}\,dx = \int \frac{1-\frac{1}{x^2}}{x^2+3+\frac{1}{x^2}}\,dx = \int \frac{(x+\frac{1}{x})'}{(x+\frac{1}{x})^2+1}\,dx = \arctan\left(x+\frac{1}{x}\right)+C.$$

Q26. $$\int \frac{1}{\sin^4 x+\cos^4 x}\,dx$$

[#1] $$= \int \frac{1}{(\sin^2 x+\cos^2 x)^2 - 2\sin^2 x\cos^2 x}\,dx$$

$$= \int \frac{1}{1-\frac{1}{2}\sin^2(2x)}\,dx = \int \frac{1}{\cos^2(2x)+\sin^2(2x)-\frac{1}{2}\sin^2(2x)}\,dx$$

$$= \int \frac{1}{\frac{1}{2}\cos^2(2x)+\frac{1}{2}}\,dx = \int \frac{\sec^2(2x)}{\frac{1}{2}+\frac{1}{2}\sec^2(2x)} = 2\int \frac{\sec^2(2x)}{2+\tan^2(2x)}\,dx$$

$\overset{t=\tan 2x}{=\!=}$ $$\int \frac{1}{2+t^2}\,dt = \frac{1}{\sqrt{2}}\arctan\left(\frac{t}{\sqrt{2}}\right)+C = \frac{1}{\sqrt{2}}\arctan\left(\frac{\tan 2x}{\sqrt{2}}\right)+C.$$

[#2] $$= \int \frac{\sec^4 x}{\tan^4 x+1}\,dx \overset{u=\tan x}{=\!=} \int \frac{1+u^2}{1+u^4}\,du = \int \frac{\frac{1}{u^2}+1}{u^2+\frac{1}{u^2}}\,du$$

$$= \int \frac{(u-\frac{1}{u})'}{(u-\frac{1}{u})^2+2}\,du = \frac{1}{\sqrt{2}}\arctan\left(\frac{u-\frac{1}{u}}{\sqrt{2}}\right)+C = \frac{1}{\sqrt{2}}\arctan\left(\frac{\tan x-\frac{1}{\tan x}}{\sqrt{2}}\right)+C.$$

Q27. $$\int \frac{\sin^2 x}{1+\sin x\cos x}\,dx$$

[#1] Let $\int \frac{\sin^2 x}{1+\sin x\cos x}\,dx =: I_1$, $\int \frac{\cos^2 x}{1+\cos x\sin x}\,dx =: I_2$.

$$I_1+I_2 = \int \frac{\sin^2 x+\cos^2 x}{1+\sin x\cos x}\,dx = \int \frac{1}{1+\sin x\cos x}\,dx = \int \frac{\sec^2 x}{\sec^2 x+\tan x}\,dx$$

$$\overset{t=\tan x}{=\!=} \int \frac{dt}{t^2+1+t} = \int \frac{1}{(t+\frac{1}{2})^2+\frac{3}{4}}\,dt = \frac{1}{\frac{\sqrt{3}}{2}}\arctan\left(\frac{t+\frac{1}{2}}{\frac{\sqrt{3}}{2}}\right) = \frac{2}{\sqrt{3}}\arctan\left(\frac{2t+1}{\sqrt{3}}\right).$$

$$= \frac{2}{\sqrt{3}}\arctan\left(\frac{2\tan x+1}{\sqrt{3}}\right) + C_1$$

$$I_1-I_2 = \int \frac{\sin^2 x-\cos^2 x}{1+\sin x\cos x}\,dx = \int \frac{-\cos(2x)}{1+\frac{1}{2}\sin(2x)}\,dx = -\ln\left|1+\tfrac{1}{2}\sin(2x)\right| + C_2$$

$$\Rightarrow I_1 = \frac{I_1+I_2+I_1-I_2}{2} = \frac{\frac{2}{\sqrt{3}}\arctan\left(\frac{2\tan x+1}{\sqrt{3}}\right) - \ln\left(1+\frac{1}{2}\sin(2x)\right)}{2} + C.$$

[#2] $$\int \frac{\sin^2 x}{1+\sin x\cos x}\,dx = \int \frac{\tan^2 x}{\sec^2 x+\tan x}\,dx \overset{t=\tan x}{=\!=} \int \frac{t^2}{(t^2+t+1)(t^2+1)}\,dt$$

$$= \int \frac{(-t)(t^2+1)+t(t^2+t+1)}{(t^2+t+1)(t^2+1)}\,dt = \int \frac{-t}{t^2+t+1}\,dt + \int \frac{t}{t^2+1}\,dt$$

$$= -\frac{1}{2}\left(\int \frac{2t+1}{t^2+t+1}\,dt - \int \frac{1}{t^2+t+1}\,dt\right) + \frac{1}{2}\ln|t^2+1|$$

$$= -\frac{1}{2}\ln|t^2+t+1| + \frac{1}{2}\cdot\frac{1}{\frac{\sqrt{3}}{2}}\arctan\left(\frac{t+\frac{1}{2}}{\frac{\sqrt{3}}{2}}\right) + \frac{1}{2}\ln|t^2+1| + C$$

$$= \frac{1}{\sqrt{3}}\arctan\left(\frac{2\tan x+1}{\sqrt{3}}\right) - \frac{1}{2}\ln\left(1+\frac{1}{2}\sin(2x)\right) + C.$$

Q28. $$\int \frac{\cos x}{\sin^2 x+\sin x+1}\,dx \overset{u=\sin x}{=\!=} \int \frac{1}{u^2+u+1}\,du = \frac{1}{\frac{\sqrt{3}}{2}}\arctan\left(\frac{u+\frac{1}{2}}{\frac{\sqrt{3}}{2}}\right) + C$$

$$= \frac{2}{\sqrt{3}}\arctan\left(\frac{2u+1}{\sqrt{3}}\right) + C = \frac{2}{\sqrt{3}}\arctan\left(\frac{2\sin x+1}{\sqrt{3}}\right) + C.$$

Q29. $$\int \frac{1}{(\cos x+\sin x)^4}\,dx$$

[#1] $$\int \frac{1}{(\cos x+\sin x)^4}\,dx = \int \frac{1}{(\cos^2 x+2\cos x\sin x+\sin^2 x)^2}\,dx$$

$$= \int \frac{1}{(1+2\sin x\cos x)^2}\,dx = \int \frac{1}{\left(1+2\cdot\tan x\cdot\frac{1}{\sec^2 x}\right)^2}\,dx = \int \frac{1}{\left(1+\frac{2\tan x}{(1+\tan^2 x)}\right)^2}dx$$

$$\overset{t=\tan x}{=\!=} \int \frac{1}{\left(1+\frac{2t}{1+t^2}\right)^2}\,\frac{1}{1+t^2}\,dt = \int \frac{(t^2+1)^2}{(t+1)^4}\cdot\frac{1}{1+t^2}\,dt$$

$$= \int \frac{t^2+1}{(t+1)^4}\,dt = \int \frac{(1+t)^2-2(1+t)+2}{(t+1)^4}\,dt$$

$$= \int \left(\frac{1}{(1+t)^2} - \frac{2}{(1+t)^3} + \frac{2}{(1+t)^4}\right)dt = -\frac{1}{1+t} + \frac{1}{(1+t)^2} - \frac{2}{3(1+t)^3} + C$$

$$= -\frac{1}{1+\tan x} + \frac{1}{(1+\tan x)^2} - \frac{2}{3(1+\tan x)^3} + C\,.$$

[#2] Notice that $\cos x+\sin x = \left(\cos x\cdot\cos\frac{\pi}{4} + \sin x\cdot\sin\frac{\pi}{4}\right)\cdot\frac{1}{\frac{\sqrt2}{2}}$

$$= \cos\left(x-\frac{\pi}{4}\right)\cdot\sqrt2$$

(Harmonic form. You should have seen this from Cal I. If you are taking waves, you should be even more familiar with it.)

$$\int \frac{1}{(\sin x+\cos x)^4}\,dx = \int \frac{1}{\left(\sqrt2\cos\left(x-\pi/4\right)\right)^4}\,dx = \frac14\int \frac{1}{\cos^4(x-\pi/4)}\,dx$$

$$\overset{u=x-\pi/4}{=\!=} \frac14\int \sec^4 u\,du = \frac14\int \sec^2 u\tan^2 u\,du + \frac14\int \sec^2 u\,du$$

$$\overset{t=\tan u}{=\!=} \frac14\int t^2\,dt + \frac14\tan u = \frac14\tan u + \frac{1}{12}\tan^3 u + C$$

$$= \frac14\tan(x-\pi/4) + \frac{1}{12}\tan^3(x-\pi/4) + C\,.$$

[#3] Recurrence. Let $f=\sin x-\cos x$, $g=\sin x+\cos x$. Then

$f'=g$, $g'=-f$, $f^2+g^2=2$.

Compute $\frac{d}{dx}\frac{f}{g^n} = \frac{f'g^n - f\,n g^{n-1}\cdot g'}{g^{2n}} = \frac{g^{n+1}+nf^2g^{n-1}}{g^{2n}}$

$$= \frac{g^{n-1}(g^2+nf^2)}{g^{2n}} = \frac{g^2+nf^2}{g^{n+1}} = \frac{g^2+n(2-g^2)}{g^{n+1}}$$

$$= \frac{(1-n)\,g^2 + 2n}{g^{n+1}} .$$

Thus if we denote $I_k := \int \frac{1}{g^k}\,dx$, then

$$\int \frac{d}{dx}\frac{f}{g^n}\,dx = \int \left(\frac{(1-n)\,g^2}{g^{n+1}} + \frac{2n}{g^{n+1}}\right) dx$$

$$\Rightarrow \quad \boxed{\frac{f}{g^n} = (1-n)\,I_{n-1} + 2n\,I_{n+1}}$$

$$\Rightarrow \quad I_{n+1} = \frac{1}{2n}\left(\frac{f}{g^n} + (n-1)\,I_{n-1}\right).$$

Thus $I_4 = \frac{1}{6}\left(\frac{f}{g^3} + 2I_2\right)$

$$= \frac{1}{6}\frac{f}{g^3} + \frac{1}{3}\cdot\frac{1}{2}\left(\frac{f}{g} + 0\cdot I_0\right)$$

$$= \frac{1}{6}\,\frac{\sin x - \cos x}{(\sin x + \cos x)^3} + \frac{1}{6}\,\frac{\sin x - \cos x}{\sin x + \cos}$$

(Equivalently, this method is equivalent as multiplying $\frac{\sin x - \cos x}{\sin x - \cos x}$ and then IBP.)

Q30. $\int_0^{\pi/4} \ln(1+\tan x)\,dx \overset{u=\pi/4-x}{=\!=\!=} \int_{\pi/4}^{0} \ln(1+\tan(\pi/4-u))\,(-du)$

$$= \int_0^{\pi/4} \ln\left(1 + \frac{1-\tan u}{1+\tan u}\right) du = \int_0^{\pi/4} (\ln 2 - \ln(1+\tan u))\,du$$

$$\Rightarrow 2\int_0^{\pi/4} \ln(1+\tan x)\,dx = \int_0^{\pi/4} \ln 2 = \frac{\pi}{4}\ln 2$$

$$\Rightarrow \quad \int_0^{\pi/4} \ln(1+\tan x)\,dx = \frac{\pi}{8}\ln 2.$$

Q31. $I = \int_0^{\pi} \frac{x\sin x}{1+\cos^2 x}\,dx \overset{u=\pi-x}{=\!=\!=} \int_{\pi}^{0} \frac{(\pi-u)\sin(\pi-u)}{1+\cos^2(\pi-u)}\,(-du)$

$$= \int_0^{\pi} \frac{\pi\sin(\pi-u)}{1+\cos^2(\pi-u)}\,du - \int_0^{\pi} \frac{u\sin(\pi-u)}{1+\cos^2(\pi-u)}\,du$$

$$= \int_0^{\pi} \frac{\pi\sin(u)}{1+(-\cos u)^2}\,du - \int_0^{\pi} \frac{u\sin(u)}{1+(-\cos u)^2}\,du$$

$\overset{t=-\cos u}{=\!=\!=} \int_{-1}^{1} \frac{\pi}{1+t^2}\,dt - I$

$\Rightarrow 2I = \pi \arctan(t)\Big|_{-1}^{1} = \pi \cdot \left[\frac{\pi}{4} - \left(-\frac{\pi}{4}\right)\right] = \frac{\pi^2}{2}$

$\Rightarrow I = \frac{\pi^2}{4}$.

Q32. $\int_0^{\pi/2} \frac{1}{1+\tan^n x}\,dx, \quad n \in \mathbb{Z}^+$

$\overset{u=\pi/2-x}{=\!=\!=} \int_{\frac{\pi}{2}}^{0} \frac{1}{1+\tan^n(\pi/2-u)}(-du) = \int_0^{\pi/2} \frac{1}{1+\left(\frac{1}{\tan u}\right)^n}\,du = \int_0^{\pi/2} \frac{\tan^n u}{\tan^n u + 1}\,du = I.$

$\Rightarrow 2I = \int_0^{\pi/2} \frac{1}{1+\tan^n x}\,dx + \int_0^{\pi/2} \frac{\tan^n x}{1+\tan^n x}\,dx = \int_0^{\pi/2} 1\,dx = \pi/2$

$\Rightarrow I = \frac{\pi}{4}$.

Q33. $\int_0^{\pi/2} \ln(\sin x)\,dx \overset{u=\pi/2-x}{=\!=\!=} \int_0^{\pi/2} \ln(\sin(\pi/2-u))\,du = \int_0^{\pi/2} \ln(\cos(u))\,du$

$\Rightarrow 2I = \int_0^{\pi/2} \ln(\sin x)\,dx + \int_0^{\pi/2} \ln(\cos(x))\,dx = \int_0^{\pi/2} \ln(\sin x \cos x)\,dx$

$= \int_0^{\pi/2} \ln\left(\frac{1}{2}\sin(2x)\right)dx = \int_0^{\pi/2} \ln(\sin(2x))\,dx - \int_0^{\pi/2} \ln 2\,dx$

$\overset{t=2x}{=\!=\!=} \frac{1}{2}\int_0^{\pi} \ln(\sin t)\,dt - \frac{\pi}{2}\ln 2$

$= \frac{1}{2}\int_0^{\pi/2} \ln(\sin t)\,dt + \frac{1}{2}\int_{\pi/2}^{\pi} \ln(\sin(\pi-t))\,dt - \frac{\pi}{2}\ln 2$

$\overset{s=\pi-t}{=\!=\!=} \frac{1}{2}I + \frac{1}{2}\int_{\pi/2}^{0} \ln(\sin(s))(-ds) - \frac{\pi}{2}\ln 2$

$= \frac{1}{2}I + \frac{1}{2}\int_0^{\pi/2} \ln(\sin(s))\,ds - \frac{\pi}{2}\ln 2 = I - \frac{\pi}{2}\ln 2$

$\Rightarrow I = -\frac{\pi}{2}\ln 2$.

Q34. $\int_0^{\pi/2} \frac{\sin^2 x}{\sin x + \cos x}\,dx \overset{u=\frac{\pi}{2}-x}{=\!=\!=} \int_{\pi/2}^{0} \frac{\sin^2(\pi/2-u)}{\sin(\frac{\pi}{2}-u)+\cos(\frac{\pi}{2}-u)}(-du)$

$= \int_0^{\pi/2} \frac{\cos^2 u}{\cos u + \sin u}\,du =: I$

$\Rightarrow 2I = \int_0^{\pi/2} \frac{\sin^2 u + \cos^2 u}{\cos u + \sin u}\,du = \int_0^{\pi/2} \frac{1}{\cos u + \sin u}\,du = \int_0^{\pi/2} \frac{1}{\sqrt{2}\cos(u-\frac{\pi}{4})}\,du$

$\overset{t=u-\pi/4}{=\!=} \int_{-\pi/4}^{\pi/4} \frac{1}{\sqrt{2}\cos t}\,dt = \left[\frac{1}{\sqrt{2}}\ln|\sec t+\tan t|\right]_{-\pi/4}^{\pi/4}$

$= \frac{1}{\sqrt{2}}\left(\ln(\sqrt{2}+1) - \ln(\sqrt{2}-1)\right)$

$= \frac{1}{\sqrt{2}}\left(\ln(\sqrt{2}+1) + \ln\left(\frac{1}{\sqrt{2}-1}\right)\right)$

$= \frac{2}{\sqrt{2}}\ln(\sqrt{2}+1)$

$\Rightarrow I = \frac{1}{\sqrt{2}}\ln(\sqrt{2}+1)$.

Q35. $\int x^3(\ln x)^2\,dx$

$u=(\ln x)^2$, $dv = x^3\,dx$

$du = \frac{2\ln x}{x}$ $\quad v = \frac{x^4}{4}$

$= (\ln x)^2\frac{x^4}{4} - \int \frac{x^3\ln x}{2}\,dx$

$= (\ln x)^2\frac{x^4}{4} - \frac{1}{2}\left(\ln x\cdot\frac{x^4}{4} - \int\frac{x^3}{4}\,dx\right)$

$u=\ln x$, $dv = x^3\,dx$

$du = \frac{1}{x}$, $v = \frac{x^4}{4}$

$= (\ln x)^2\frac{x^4}{4} - \frac{1}{2}\left(\ln x\cdot\frac{x^4}{4} + \frac{x^4}{16}\right) + C$

Hence $\int_0^1 x^3(\ln x)^2\,dx = \lim_{t\to 0}\int_t^1 x^3(\ln x)^2\,dx$

$= \lim_{t\to 0}\left[\ln(x)^2\frac{x^4}{4} - \ln x\cdot\frac{x^4}{8} + \frac{x^4}{32}\right]_t^1$

$= 0 - 0 + \frac{1}{32} - 0 + 0 - 0 = \frac{1}{32}$.

Q36. $\int \frac{x^2}{(x^2+1)^2}\,dx \overset{x=\tan\theta}{=\!=} \int \frac{\tan^2\theta}{\sec^4\theta}\sec^2\theta\,d\theta = \int\frac{\tan^2\theta}{\sec^2\theta}\,d\theta = \int\sin^2\theta\,d\theta$

$= \int\frac{1-\cos(2\theta)}{2}\,d\theta = \frac{1}{2}\theta - \frac{\sin(2\theta)}{4} + C = \frac{1}{2}\arctan x - \frac{\sin(2\arctan x)}{4} + C.$

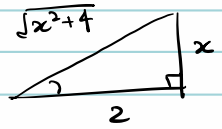
Thus $\int_0^\infty \frac{x^2}{(x^2+1)^2}\,dx = \lim_{t\to\infty}\left[\frac{1}{2}\arctan x - \frac{\sin(2\arctan x)}{4}\right]_0^t$

$= \frac{\pi}{4} - 0 - 0 + 0 = \pi/4.$

Q37. $\int \frac{1}{(x^2+1)\sqrt{x^2+4}}\,dx \overset{x=2\tan\theta}{=\!=\!=} \int \frac{2\sec^2\theta\,d\theta}{(4\tan^2\theta+1)\,2\sec\theta} = \int \frac{\sec\theta}{4\tan^2\theta+1}\,d\theta$

$= \int \frac{\sec\theta}{4\frac{\sin^2\theta}{\cos^2\theta}+1}\,d\theta = \int \frac{\cos\theta}{4\sin^2\theta+\cos^2\theta}\,d\theta = \int \frac{\cos\theta}{3\sin^2\theta+1}\,d\theta \overset{u=\sin\theta}{=\!=\!=} \int \frac{1}{3u^2+1}\,du$

$= \frac{\sqrt{3}}{3}\arctan(\sqrt{3}u) + C = \frac{\sqrt{3}}{3}\arctan\left(\sqrt{3}\sin\left(\arctan\left(\frac{x}{2}\right)\right)\right) + C$



$= \frac{\sqrt{3}}{3}\arctan\left(\sqrt{3}\frac{x}{\sqrt{x^2+4}}\right) + C$

Q38. $\int \frac{x^2\arctan x}{1+x^2}\,dx = \int \arctan x\,dx - \int \frac{\arctan x}{1+x^2}\,dx$

$= \left(x\arctan x - \int \frac{x}{1+x^2}\,dx\right) - \frac{1}{2}(\arctan x)^2$

$= x\arctan x - \frac{1}{2}\ln(1+x^2) - \frac{1}{2}(\arctan x)^2 + C.$

Q39. $\int \sqrt{2\sqrt{x}-x}\,dx \quad [\#1] = \int \sqrt[4]{x}\sqrt{2-\sqrt{x}}\,dx$

$\overset{x=4\sin^4\theta}{=\!=\!=} \int \sqrt{2}\sin\theta\,\sqrt{2}\cos\theta\cdot 16\sin^3\theta\cdot\cos\theta\,d\theta$

$= \int 32\sin^4\theta\cos^2\theta\,d\theta$

$= 32\int \left(\frac{1-\cos(2\theta)}{2}\right)^2\left(\frac{\cos 2\theta+1}{2}\right)d\theta$

$= 4\int (\cos^3 2\theta - \cos^2 2\theta - \cos 2\theta + 1)\,d\theta$

$= 4\int \left[(1-\sin^2 2\theta)\cos 2\theta - \cos^2 2\theta - \cos 2\theta + 1\right]d\theta$

$= -\frac{2}{3}\sin^3 2\theta - 4\int \frac{1+\cos 4\theta}{2}\,d\theta + 4\int d\theta$

$= -\frac{2}{3}\sin^3 2\theta + 2\theta - \frac{\sin(4\theta)}{2} + C$

$= -\frac{2}{3}\sin^3\left(2\arcsin\left(\sqrt[4]{\frac{x}{4}}\right)\right) + 2\arcsin\left(\sqrt[4]{\frac{x}{4}}\right) - \frac{\sin\left(4\arcsin\left(\sqrt[4]{\frac{x}{4}}\right)\right)}{2} + C$

[#2] $\int \sqrt{2\sqrt{x} - x}\, dx \overset{u=\sqrt{x}}{=\!=\!=} \int \sqrt{2u - u^2} \cdot 2u\, du$

$= \int 2u\sqrt{1-(u-1)^2}\, du \overset{u-1=\sin\theta}{=\!=\!=} \int 2(\sin\theta+1)\cos\theta\cos\theta\, d\theta$

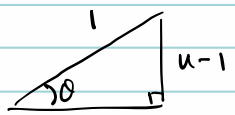
$= \int 2\sin\theta\cos^2\theta\, d\theta + \int 2\cos^2\theta\, d\theta$

$= -\frac{2}{3}\cos^3\theta + 2\int \frac{1+\cos 2\theta}{2}\, d\theta$

$= -\frac{2}{3}\cos^3\theta + \theta + \frac{\sin 2\theta}{2} + C$

$= -\frac{2}{3}\cos^3(\arcsin(u-1)) + \arcsin(u-1)$

$+ \sin(\arcsin(u-1))\cos(\arcsin(u-1)) + C$



$\sqrt{1-(u-1)^2} = \sqrt{2u-u^2} = \sqrt{2\sqrt{x}-x}$

$= -\frac{2}{3}\left(\sqrt{2\sqrt{x}-x}\right)^3 + \arcsin(\sqrt{x}-1) + (\sqrt{x}-1)\sqrt{2\sqrt{x}-x} + C$